

Local Gain–Loss Click Sequences at a Hard Controller Domain Wall

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We specify a trajectory-resolved pilot for local gain and loss records in a number-conserving adaptive Gaussian circuit with a domain-wall-truncated controller. The simulation uses the canonical CPU dynamics entry point and records, without covariance histories, the four measurement-feedback channels generated by two Pauli trial eigenvectors and two flattened Hamiltonian bands. We define a four-bit unit-cell motif, temporal and periodic spatial words, shuffle null ensembles, and trajectory-bootstrap uncertainty. Two matched ten-trajectory schedules, a deterministic y -raster and a randomized site order, separate schedule-conditioned patterns from candidates that persist under update-order randomization. This document is a methods specification: it states estimators, validation gates, and interpretive limits but makes no claim about the numerical outcome of the pilot.

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I. GAUSSIAN INSTRUMENT AND TRIAL EIGENVECTORS

The physical single-particle covariance is

$$G = 2C - \mathbf{1}, \quad C_{ij} = \text{Tr}(\rho c_i^\dagger c_j), \quad (1)$$

with two complex orbitals per unit cell. At a controller center $R = (R_x, R_y)$, choose a Pauli operator σ_{trial} . Its normalized eigenvectors are denoted

$$\sigma_{\text{trial}}\tau_A = +\tau_A, \quad \sigma_{\text{trial}}\tau_B = -\tau_B. \quad (2)$$

Thus A/B are the two eigenvectors of the chosen Pauli operator, not merely post hoc channel names. The pilot fixes $\sigma_{\text{trial}} = \sigma_x$, or `trial_orbitals="X"`. Projection of each trial eigenvector into the upper and lower flattened-Hamiltonian bands gives four normalized observable Wannier modes,

$$|w_{A+;R}\rangle, \quad |w_{A-;R}\rangle, \quad |w_{B+;R}\rangle, \quad |w_{B-;R}\rangle. \quad (3)$$

The target occupations are

$$s_{A+} = s_{B+} = 0, \quad s_{A-} = s_{B-} = 1. \quad (4)$$

The implemented order inside a visited cell is $A+, A-, B+, B-$. Because a measurement changes the covariance before the next channel is sampled, the four outcomes at one cell are an ordered local instrument record.

II. HARD-WALL CONTROLLER GEOMETRY

The simulation fixes

$$N_x = 20, \quad N_y = 24, \quad n_{\text{shell}} = 1, \quad \alpha_1 = 1, \quad \alpha_2 = 30, \quad (5)$$

and constructs a topological slab using `DW=True`. The Wannier support is first cut to $n_{\text{shell}} = 1$ and then masked according to which side of the controller domain wall contains its center:

$$|w_{a;R}^{\text{DW}}\rangle = \frac{M_R |w_{a;R}\rangle}{\|M_R |w_{a;R}\rangle\|}. \quad (6)$$

The flags `dw_truncation=True` and `meas_slab_only=True` are mandatory. For $N_x = 20$, the class rule places the active slab at $4 \leq x \leq 16$, with hard walls at $x = 4, 16$. The interface mask has width $n_{\text{shell}} + 1 = 2$, so the left and right interface columns are $\{4, 5\}$ and $\{15, 16\}$; columns $6, \dots, 14$ form the active interior control.

Every trajectory is initialized with `init_mode="maxmix"` and evolved for 48 complete sweeps. Cycles $1, \dots, 24$ are burn-in and cycles $25, \dots, 48$ define the observation window. No covariance history is saved. Production scripts call only

$$\text{classA.U1FGTN.run_markov_circuit}(\dots) \quad (7)$$

with `G_history=False` and `save=False`; the branch observer streams the classical record before the final covariance is discarded.

III. SIGNED CLICK RECORD AND FOUR-BIT MOTIFS

For channel a at the m th visited cell, let $n_{a,m} \in \{0, 1\}$ be the Born-rule measurement and $n'_{a,m} \in \{0, 1\}$ the occupation after conditional feedback. We retain both the target defect

$$X_{a,m} = |n_{a,m} - s_a| \quad (8)$$

and signed particle transfer

$$Y_{a,m} = n'_{a,m} - n_{a,m} \in \{-1, 0, +1\}. \quad (9)$$

Negative, zero, and positive Y denote loss, no transfer, and gain. Perfect correction is used, hence

$$X_{a,m} = |Y_{a,m}| \quad (10)$$

event by event. Equation (4) then restricts the four nonzero click types as follows:

TABLE I. Four click bits retained at each unit-cell visit.

Bit	Controller channel	Trial eigenvector	Signed event
0	$A+$	τ_A , eigenvalue +1 of σ_x	A-loss, $Y = -1$
1	$A-$	τ_A , eigenvalue +1 of σ_x	A-gain, $Y = +1$
2	$B+$	τ_B , eigenvalue -1 of σ_x	B-loss, $Y = -1$
3	$B-$	τ_B , eigenvalue -1 of σ_x	B-gain, $Y = +1$

Define $b_j \in \{0, 1\}$ from these four events and encode the cell motif as

$$q = b_0 + 2b_1 + 4b_2 + 8b_3 \in \{0, \dots, 15\}. \quad (11)$$

This representation is deliberately not a single trinary A/B value. A cell can exhibit both A-loss and A-gain in one ordered visit; Eq. (11) preserves both bits instead of cancelling them. Motif $q = 0$ is the all-no-click state.

Each channel row stores the schedule, sample, one-based cycle, within-cycle cell visit, zero-based global cell and channel steps, x, y , interface label, distances to both walls, A/B trial eigenvector, Pauli eigenvalue, band, target, X, Y , click label, Born success probability, and validity. A second table stores one row per unit-cell visit with q and all four bits. The compressed array record remains the source of truth.

IV. TEMPORAL AND SPATIAL WORD STATISTICS

For a fixed cell R , temporal words of length $k = 2, 3$ are

$$W_k^{(t)}(R, t) = (q(R, t), q(R, t + 1), \dots, q(R, t + k - 1)). \quad (12)$$

Words never cross a trajectory, cell, or burn-in boundary. At fixed (x, t) , spatial words follow the positive periodic y direction:

$$W_k^{(y)}(x, y, t) = (q(x, y, t), q(x, y + 1, t), \dots, q(x, y + k - 1, t)), \quad y \equiv y + N_y. \quad (13)$$

Counts are reported for the left wall, right wall, pooled interface, active interior, and entire active slab.

The temporal null independently permutes the post-burn-in cycle labels for every trajectory and cell. It therefore preserves the one-point motif count of each cell while destroying temporal order. The spatial null independently permutes motif positions around each (trajectory, t, x) periodic y -ring. It preserves the ring's motif multiset while destroying nearest-neighbor order. Both null ensembles contain 1000 permutations generated from a recorded analysis seed.

For observed word count N_w , total count N , and a word space of size $K = 16^k$, the Jeffreys-regularized frequency is

$$\hat{f}_w = \frac{N_w + \frac{1}{2}}{N + \frac{1}{2}K}. \quad (14)$$

The reported effect size is

$$E_w = \log_2 \frac{\hat{f}_w}{\langle \hat{f}_w^{\text{null}} \rangle}. \quad (15)$$

Uncertainty is obtained from 1000 trajectory-level bootstrap resamples paired with null-permutation replicates. A ranked table retains words with at least 20 observations. Its exploratory candidate flag only indicates that the percentile interval for Eq. (15) excludes zero; it is not a multiple-testing-corrected discovery statement.

V. MATCHED SCHEDULE DESIGN AND DEPLOYMENT

Two ten-trajectory ensembles use a common root seed 20260814 and identical derived sample seeds. The first uses **raster-y**, which visits columns left-to-right and sites within each column in increasing y . Only after that campaign succeeds does the launcher begin the **random** schedule, which reshuffles the active sites every cycle. The schedules therefore form a matched update-order comparison, not concurrently competing jobs.

Within a schedule, ten independent trajectories run on ten single-thread process workers. BLAS, OpenMP, MKL, and NumExpr thread counts are fixed to one. The launcher creates a new timestamped directory, refuses overwrites, writes separate stage logs, and uses **_SUCCESS** or **_FAILED** markers. If the pre-existing CFT campaign remains active on CPUs 0–55, the pilot is pinned to CPUs 56–111. The all-core topological-frustration campaign is allowed to finish before production begins and is never interrupted.

VI. VALIDATION GATES

A production stage is valid only if:

1. every active cell is visited once per cycle and each visit contains exactly the four expected channel measurements;
2. all valid Born success probabilities are finite and lie in $[0, 1]$;
3. $A+, B+$ transfers belong to $\{0, -1\}$, while $A-, B-$ transfers belong to $\{0, +1\}$;
4. Eq. (10) holds elementwise;
5. dense-array dimensions agree with both tidy-table row counts;
6. all geometry and controller parameters in Eq. (5), the Pauli basis, canonical dynamics entry point, seeds, and CPU policy are present in metadata;

7. raster and random stages contain identical initialization and sample seed lists.

Unit tests exhaust all 16 motif codes, simultaneous gain/loss events, temporal boundary handling, periodic- y closure, shuffle count preservation, determinism, and malformed records. A 4×6 , two-sample, two-cycle run validates both schedules and the full artifact schema before production.

VII. INTERPRETIVE LIMITS

This is a ten-trajectory pilot designed to identify candidate effects, not thermodynamic or long-time laws. A raster-only spatial word may be generated by the update schedule itself. Persistence under random ordering is a stronger robustness criterion, but does not by itself establish universality. The hard mask is a boundary of the controller support, not a smooth material interface. Perfect correction also identifies wrong outcomes with absolute transfer activity and therefore does not test failed feedback. These limits must accompany every numerical summary derived from the pilot.